

CS-552 Course Project

Spring 2026 — Standardized Project

Modern Natural Language Processing

Project Overview

Welcome to the CS-552 course project! If you select the **standardized NLP research project** option, you will find important information below to help you understand the important steps and deliverables.

In this project, you will work as a team of 4 students to train large reasoning models capable of solving different downstream reasoning problems. Starting from **the same base model**, each team will apply **post-training methods** of their choice to produce a model that can generate correct, verifiable solutions to questions from four areas: **Mathematics, General Knowledge, Multilinguality, and Safety**. This project offers hands-on experience with modern post-training techniques that are central to building reasoning-capable large language models.

Table of Contents

Project Overview.....	1
Task Description.....	2
Base Model & Infrastructure.....	3
Timeline & Deliverables.....	4
Part I — Proposal (10%), Due on May 3.....	5
Deliverables.....	5
2-Page Report.....	5
Part II — Model Running Validation (10%), Due on May 24.....	6
Deliverables.....	6
1. Group Model (1%).....	6
2. Individual Model (9%).....	6
Part III — Final Submission (50%), Due June 7.....	7
Deliverables.....	7
1. Final Report (30%).....	7
2. Individual Model (15%).....	8
2. Group Model (5%).....	8
Important Information.....	8
Mentorship.....	8
Collaboration.....	8
Compute Resources.....	8
Submission.....	8

Task Description

Each team will train a reasoning model using one or more post-training methods, including but not limited to:

- Supervised Fine-Tuning (SFT)
- Reinforcement Learning from Human Feedback (RLHF)
- Reinforcement Learning with Verifiable Rewards (RLVR)
- Any combination of the above methods
- Or other methods that you want to try

Teams may choose to train models that generate a thinking trajectory before producing the final answer (similar to DeepSeek-R1: <https://arxiv.org/abs/2501.12948>), or use a standard chain-of-thought generation approach. Regardless of the approach chosen, the model's output must support automated verification: **the final answer must be enclosed in a LaTeX `\boxed{}` environment (e.g., `\boxed{42}`)** so that the evaluation suite can extract and verify it.

Team Effort vs. Individual Effort

This project is graded on both team effort and individual effort. Each team of 4 students must submit a total of 5 models: 1 group model and 4 individual models. Each student is personally responsible for training and submitting one individual model.

The four individual models target four different evaluation benchmarks, each focusing on a distinct skill:

1. **Mathematical Reasoning** — Solving math problems across various domains (algebra, geometry, number theory, combinatorics).
2. **General Knowledge** — Answering factual questions spanning world knowledge, science, history, and common sense.
3. **Multilinguality** — Reasoning faithfully in multiple languages beyond English.
4. **Safety** — Generating responses that avoid harmful, toxic, or unsafe content while adhering to ethical guidelines and responsible AI practices.

Group members decide among themselves which benchmark each person will pursue. Each individual model is evaluated only on its dedicated benchmark and is ranked against other individual models that are also trained specifically for that same benchmark. The group model, on the other hand, is evaluated on all four benchmarks and is expected to achieve uniformly good performance across all of them. This creates a natural tension: the team must balance specialization in individual models with generalization in the joint group model.

Validation Data

Students are responsible for curating their own validation evaluation data to monitor their model's intermediate performance throughout the project. This validation data should be representative of the target benchmark and should help you diagnose weaknesses, track progress, and guide iterative improvements to your training pipeline. You are free to use publicly available datasets, construct your own evaluation examples, or combine both

approaches. Thoughtful selection of validation data is a critical part of the training process and will directly impact the quality of your final model.

Base Model & Infrastructure

All teams will start from the same base model:

- Qwen3-1.7B ([Qwen/Qwen3-1.7B · Hugging Face](#))

The final trained models must be uploaded to the [Hugging Face Hub](#) as public model checkpoints that the TAs can download for grading. The models must be compatible with [vLLM](#) inference under the following predefined generation parameters:

- Max new tokens: 16384
- Seed: 42
- Stop: End-of-sequence (EoS) token
- Chat template: Applied automatically by vLLM

You can, however, modify the default sampling parameters from Qwen3's generation config, including the temperature, top_k, and top_p:

```
{
  "bos_token_id": 151643,
  "do_sample": true,
  "eos_token_id": [
    151645,
    151643
  ],
  "pad_token_id": 151643,
  "temperature": [DEFINED BY YOU],
  "top_k": [DEFINED BY YOU],
  "top_p": [DEFINED BY YOU],
  "transformers_version": "4.51.0"
}
```

When submitting your model, make sure this **generation config** is submitted as part of the **model checkpoint** (e.g., following the [structure of the Qwen3 model checkpoint](#) on Huggingface Hub). You must ensure that the submitted generation config in your checkpoint is exactly the one you intend to use for the TAs to grade your model performance.

Models will be evaluated using the [OpenCompass](#) framework, which automatically extracts the final answer from the `\boxed{}` pattern in the model generations and compares against gold solutions. Teams are evaluated on the accuracy of their model's answers across the evaluation benchmarks.

Timeline & Deliverables

The project (which is worth 70% of your grade) is divided into three milestones:

Milestone	Weight	Due Date	Deliverables
Proposal	10%	May 3, 2026	1-page proposal (5%) + 1-page literature review (5%)
Model Running Validation	10%	May 24, 2026	group model (1%) + individual model (9%)
Final Submission	50%	June 7, 2026	4-page report (30%) + group model (5%) + individual model (15%)

Part I — Proposal (10%), Due on May 3

The proposal is your opportunity to define the scope and direction of your project. A clear, well-motivated proposal will set the foundation for a successful project. The proposal should outline the post-training method(s) you intend to use (SFT, RLHF, RLVR, or a combination), your training data strategy, evaluation data & strategies for you to validate your models' performance, and any preliminary experiments or baselines you plan to establish.

Deliverables

1 Page Proposal

Submit a short report (maximum of 1 page, excluding references) outlining your project plan. The proposal should be submitted jointly by all team members and should contain the following sections.

Introduction. Introduce your project idea concisely. What problem are you addressing? What is the task or research question? Clearly state the goal of the project.

Methodology. Describe your execution plan in detail. What post-training methods do you intend to use? What datasets will you rely on, and how will you access or preprocess them? What evaluation metrics will you use?

Project Timeline. Outline a realistic timeline for the remainder of the project, with concrete goals for the Milestone and Final Submission deadlines. Additionally, please include which team member will contribute to what part of the project.

You must use the provided [LaTeX template](#) for the proposal.

1 Page Literature Review.

Provide a brief survey of the relevant literature (maximum of 1 page, excluding references). What prior work exists on this topic? How does your proposed approach relate to or differ from existing methods? This section should justify why your project is needed and contextualize it within the broader research landscape.

A good literature review synthesizes themes across papers rather than simply summarizing them one by one and groups related work by topic, method or perspective. We recommend you check out the related works section of papers published in top conferences (ACL, EMNLP, NAACL, NeurIPS, ICLR, ICML, AAAI) for examples of good literature review. You can also follow [this guideline](#) to effectively read and review papers in general.

Completing this review will teach you how to (1) find research papers related to a topic you are studying, (2) critically assess and compare research papers, and (3) gather useful information from them that you can re-implement in your own work.

You must use the provided [LaTeX template](#) for the literature review.

Rubric

Proposal (5%)

1. Introduction: Problem clarity and motivation for the plan discussed (**1 point**)
2. Methodology: Technical Plan (eg, the post-training method chosen), training data, validation datasets and evaluation strategy (**3 points**)
3. Project timeline and team roles (**1 point**)

Related Work (5%)

1. 12+ references: You need to refer to at least 12 research papers (i.e., 3 per person) (**2 points**)
2. Contextualization and coherent synthesis of these works. (**3 points**)
3. Any detected hallucinations in the references get you a 0.

Part II – Model Running Validation (10%), Due on May 24

A continuous integration (CI) pipeline runs on a dedicated GPU node to automatically validate all submitted models. The pipeline periodically checks the Hugging Face Hub for updated model checkpoints at the end of each day (23:00 p.m.) — both group models and individual models — by monitoring commit timestamps. When an update is detected, the pipeline automatically performs the following checks:

1. Model loading: Verifies that the model can be successfully loaded by the vLLM inference engine.
2. Inference: Runs the model on 5 validation examples to confirm it produces meaningful generations.
3. Answer extraction: Checks that the evaluation pipeline can successfully extract a verifiable final answer from each generation using the required `\boxed{ANSWER}` syntax.

Results and failure reports from each validation run will be distributed to the students, so you can diagnose and fix issues during development. You may continue to update your models on the Hugging Face Hub at any time. The CI pipeline will automatically pick up and re-validate new commits until the final due date.

To receive the full 10% for this milestone, both your group model and all individual models must pass the model-running and generation validation checks by the due date.

Deliverables

1. Group Model (1%)

Upload your team's group model checkpoint to Hugging Face Hub. The CI pipeline will verify that the model loads correctly with vLLM, produces meaningful generations on the validation examples, and outputs answers in the required `\boxed{}` format.

2. Individual Model (9%)

Each team member must also upload their own individual model checkpoint to Hugging Face Hub. The same CI pipeline validates each individual model independently. Both your group model and your individual model must pass all validation checks for you to receive the full 10% milestone grade.

Rubric

Will be released with the milestone.

Part III — Final Submission (50%), Due June 7

The final submission is the culmination of your project. Each team must submit a total of 5 model checkpoints to the Hugging Face Hub: 1 group model and 4 individual models (one per team member). The submission also includes a research-style report.

Deliverables

1. Final Report (30%)

Submit a report (maximum of 4 pages, excluding references and appendices) written in the style of an NLP or machine learning research paper. You must use the provided LaTeX template. The report should include the following sections.

Abstract. A concise summary (under 300 words) that motivates the problem, describes your approach and contribution, and highlights the main findings.

Introduction. Explain the problem, why it is important or interesting, how current methods succeed or fall short, and provide a high-level overview of your approach and key results.

Methodology. Describe your approach in detail, including model architecture, training procedure, data pipeline, and any other key methods. Include equations and figures where appropriate. Use consistent, well-defined technical vocabulary throughout. Clearly distinguish between the training strategy for the group model and the individual models, and explain how you balanced generalization (group) with specialization (individual).

Experiments. This section should cover the following sub-sections:

- **Data:** Describe the dataset(s) you used, including sources, preprocessing, and any legal or ethical considerations.
- **Evaluation:** Describe the evaluation metrics and methodology, including any baselines you compare against.
- **Experimental details:** Report model configurations, hyperparameters, training time, hardware used, and any other details needed for reproducibility.
- **Results:** Present **empirical quantitative** evaluation results using tables and/or plots. Compare against baseline results separately for each individual model on its dedicated benchmark (mathematical reasoning, general knowledge, multilinguality, safety) and for the group model across all four benchmarks. Report major findings and contributions based on the results. Analyze any interesting trends and report associated findings and takeaways.

Ethical Considerations. Discuss the broader impact of your work. You should address when applicable: how the model could be adapted to other languages (high-resource and low-resource); how it might be adapted for signed-language interaction; who benefits and who might be harmed; and whether any harms disproportionately affect marginalized groups.

Conclusion. Summarize your main findings, highlight key achievements, note primary limitations, and optionally suggest directions for future work.

2. Individual Model (15%)

Each team member must upload their individual model checkpoint to the Hugging Face Hub. Each individual model is trained and evaluated on a single dedicated benchmark (mathematical reasoning, general knowledge, multilinguality, or safety) and is ranked against other individual models targeting the same benchmark. The individual model grade reflects each student's personal contribution and is not shared with other team members. Your checkpoint must include the generation config and be compatible with vLLM inference.

2. Group Model (5%)

Upload your team's final group model checkpoint to the Hugging Face Hub. The group model is evaluated on all four benchmarks (mathematical reasoning, general knowledge, multilinguality, and safety) and is expected to achieve uniformly good performance across all of them. The group model grade is shared equally among all team members. Your checkpoint must include the generation config and be compatible with vLLM inference.

Rubric

Will be released with the milestone.

Important Information

Mentorship

Each team will be mentored by a course AE. Your mentor is your first point of contact for project-related questions. Mentors are available for discussion during normal course hours. Alternative meeting times, including remote meetings, can be arranged as needed.

Collaboration

You must work in teams of **four**. All team members should contribute roughly equally. You are free to discuss your project with others in the course, but only team members should contribute to the implementation and experimentation.

Compute Resources

All teams have access to a GPU cluster for training and evaluation. Refer to the course documentation for details on accessing these resources.

Submission

All submissions are made through **HuggingFace**. Ensure that your repository is well-organized and includes all the important components that the original base model checkpoint contains.